

BSV EquiCare — Technical Requirements

Remaining sensor and infrastructure scope (monitoring points 1, 5, 6, 7, 8, 9, 10)

Bharat Sports Venture — EquiCare engineering · September 2026

1. What we are building

A 24/7 monitoring system for stabled horses in India. Each stall is instrumented; sensors feed an **on-site edge computer (NVIDIA Jetson, ARM64)**, and our software turns raw data into health and behaviour insight.

We do our own analytics. We are not buying algorithms, dashboards or scoring engines. We need **raw, timestamped, per-sensor data** and a documented way to read it on our own hardware. Where a device computes its own classifications, those must be **disableable**, and the raw stream must remain available regardless.

Every reading, from every sensor, is normalised into one record shape:

```
{ horseId, stallId, metric, value, unit, ts, source, confidence, meta }
```

If a device can be mapped onto that, it integrates. If it can only emit a proprietary summary score, it cannot.

2. Scope of this document

The thermal camera is procured and integrated. It measures body temperature and respiratory rate directly today. This document covers **everything else**.

Monitoring point	Sensor track
1 — Steps / locomotion	IMU leg tag
5 — Activity	IMU (fused with optical)
6 — Rest / lying time	IMU (fused with optical)
7 — Lameness / limb-favouring	IMU leg tag
8 — Stable vices (weaving, crib-biting, wind-sucking)	Microphone (+ optical)
9 — Watering: quantity, pattern, time	Water metering
10 — Feeding: quantity, pattern, time	Feed metering
<i>all 12</i>	Edge compute, network, power, enclosure

3. Requirements that apply to every supplier

These are pass/fail and precede any product-specific discussion. A supplier who cannot meet them cannot be integrated, regardless of how good the sensor is.

#	Requirement	Why
G1	Linux ARM64 (aarch64) support. Any SDK, driver or library must be <i>built and tested</i> for aarch64 — not x86-64 only, not Windows/.NET only. An open protocol (Modbus, BLE GATT, ALSA, MQTT, serial) makes this a non-issue.	Our edge box is a Jetson. A previous supplier confirmed "Linux SDK — yes" and shipped an x86-64-only package with no ARM64 build available. It cost both sides weeks.
G2	Raw time-series accessible — not only vendor-computed summaries or scores.	Our gait, behaviour and acoustic models need the underlying waveform. A closed device returning an "activity score" cannot be used.
G3	Any on-device analytics can be fully disabled.	We run inference ourselves and must not have firmware silently pre-filtering the input.
G4	No mandatory vendor cloud. All data must be readable on the local network, with the entire path capable of running on-premise.	India DPDP-2023 data residency, and barns with unreliable rural internet.
G5	Documented open protocol or API, supplied as an actual file — register map, protocol spec, API reference.	We integrate directly. A capability we cannot verify against a document counts as unverified.
G6	Timestamping and clock discipline. State whether records are stamped at the device or on receipt, the sync method (PTP preferred, NTP acceptable), and worst-case drift over 24 h.	IMU, video, thermal ROI temperatures and audio must align on one timeline. We are targeting tens of milliseconds.
G7	≥24 h local buffering with automatic backfill using original timestamps , and acknowledged delivery.	Barns lose power and connectivity routinely. Data replayed with the wrong timestamps is worse than no data.
G8	Per-reading quality / confidence indicator — validity, saturation, settled-vs-moving, clip flag, or equivalent.	It carries into our data model and stops a bad sample being treated as a clinical finding.
G9	India regulatory compliance — BIS, and WPC/ETA radio approval for anything that transmits.	Legal deployment requirement.
G10	Stable, documented data schema with units , versioned, with a real sample payload.	Field names and units must be mappable to the record shape in §1.

4. IMU leg wearable — points 1, 5, 6, 7

A leg-mounted tag (cannon bone) streaming raw motion data to the edge box. All step counting, gait analysis, lameness detection and lying detection are done by us.

SENSING

Parameter	Requirement
Axes	6-axis minimum (3-axis accel + 3-axis gyro); 9-axis preferred
Accelerometer range	$\geq \pm 16 \text{ g}$ — a horse kicking or striking the stall wall clips lower ranges and corrupts exactly the data we need
Gyroscope range	$\geq \pm 2000 \text{ }^\circ/\text{s}$ — no clipping at trot, canter or kick
Output data rate	$\geq 100 \text{ Hz}$, target 100–200 Hz , configurable, and achievable <i>while streaming raw</i>
Noise	Noise density / RMS noise for accel and gyro at target ODR; any on-chip filtering must be bypassable
Temperature stability	Gyro bias drift across 45–50 °C , and field recalibration without removing the tag from the horse

State sensor part numbers and attach datasheets. On-firmware activity classification (standing / lying / walking / trotting), if offered, must be declared as equine-validated or generic, and must be disableable. Step counting must be stated as equine-validated or adapted from human/cattle, with error at walk and at trot.

DATA PATH

- Transport (BLE GATT, Wi-Fi, proprietary radio → gateway, USB, serial), with the protocol document or register map attached.
- **Continuous streaming at $\geq 100 \text{ Hz}$** , or batch upload — if batched, state interval, payload format and throughput.
- If BLE: version, **Coded-PHY / Long-Range** support, TX power, and realistic range **through masonry and steel barn structure** — not open-field figures.
- If a gateway is required: model, **tags per gateway at raw sample rate**, uplink (Ethernet/PoE preferred), and a confirmed local-only path to our edge box.

POWER, FIT AND WELFARE

- **Battery life** at each mode (lying-detection only vs 100–200 Hz raw streaming); chemistry, mAh, charge time, cycle life.
- Charging operations for a whole yard: multi-bay dock, hot-swap spares, how data gaps during a swap are handled, and how a replacement tag is **re-bound to the same horse**.
- **IP67 or better**, validated against wash-down, mud, urine/manure and sweat; operating range must cover 45 °C+ and monsoon humidity.

- Attachment to the cannon bone, size range **pony** → **draft**, strap service life.
 - **Weight target** ≤50 g, with any 24/7 wear assessment showing no rub sores.
 - **Tamper / detach alert**, and observed tag-loss rate per tag-year in the field.
 - **Globally unique immutable hardware ID** in every record, and a queryable API to bind tag → horseld.
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5. Feed and water metering — points 9 and 10

WATER MEASUREMENT

- **Measurement principle** — electromagnetic, ultrasonic, hall-effect, turbine inline, or load cell — and which is recommended for equine use.
- **Resolution and accuracy** — flow meter: **ml per pulse** and accuracy across the range.
- **Minimum measurable flow / cut-off**, in L/min and ml/min. A horse taking slow sips must register; a cut-off tuned for plumbing will miss it entirely.
- **Indian water conditions** — tolerance to **high-TDS groundwater and scaling**, plus hay and chaff fouling the vessel.
- **Refill vs. consumption** — the device must distinguish water *drunk* from water *added*.
- If load-cell based: can it separate **drinking from spillage, evaporation and play**, and what is the sampling rate and debounce? Horses nudge, lean on and bite buckets.

FEED MEASUREMENT

- **Dose resolution and accuracy in grams**, stated separately for **pellets, textured / sweet feed**, and **mash**.
- **Weigh-back / refusal** — the system must measure what the horse **actually ate** (offered minus refused), not only what was dispensed.
- **Forage / hay** — is a load-cell manger or haynet offered? State resolution.
- **Programmable per-horse schedule** (morning / midday / evening / free-choice), updatable remotely.
- **Dispense confirmation** — each dispense should emit a record with scheduled vs. actual.
- **Fault reporting** — hopper empty, bridging / jam, and motor stall must be reported as data, not left silent.

ATTRIBUTION

Per-horse attribution is mandatory (G-level). Either a dedicated per-stall vessel, or RFID identification at a shared vessel. Per-horse trends are the entire clinical value; a shared trough with no animal ID is unusable to us.

If RFID: state the standard — **LF 134.2 kHz ISO equine microchip**, UHF, or BLE — and the read range and reliability while the animal is drinking or eating.

INTEGRATION AND OPERATIONS

- **Interfaces** — Modbus RTU/RS-485, Modbus TCP, pulse / dry-contact, 4–20 mA, BLE, MQTT.

- **Event model** — continuous totals, or discrete bouts with start and end times? We need pattern and timing, not only daily volume.
- **Devices per edge box** — RS-485 addressing limits and throughput when ingesting water and feed across a barn.
- **Calibration** — procedure, frequency, whether barn staff can do it in the field; documented drift spec (zero and span, over temperature and time) and auto-tare behaviour.
- **Mechanical abuse rating** — kick, lean, crib, rub. Is a guard or recessed mount offered?
- **Hygiene** — water- and feed-contact parts must be removable and safe for daily disinfectant wash-down; **IP66/IP67 on housing *and* connectors/glands**, corrosion resistant, **food/potable-grade wetted materials**.
- **Power** — wired DC, PoE (state class and watts), mains or battery. PoE shares the barn power budget in §7.
- **Vessel capacity** (litres / kg), and whether a manual refill can be logged as such.
- **Remote firmware and config update** from the edge, without a site visit.

6. Microphone — point 8, and cough detection supporting 3–4

Stall audio for stable-vice detection (weaving, crib-biting, wind-sucking) and respiratory sound. All classification is ours.

THE TWO REQUIREMENTS THAT DECIDE EVERYTHING

#	Requirement	Why
M1	Uncompressed PCM (or lossless) audio available to us — not only AAC, Opus or G.711.	This is the single most important item in this section. Lossy voice codecs destroy exactly the transients we classify on: teeth-on-wood contact, the onset of a wind-suck, the sharp front of a cough.
M2	AGC, noise suppression, echo cancellation and any noise gate can be FULLY DISABLED , in favour of fixed or manual gain.	AGC flattens the level information that separates a cough from a door bang, and a noise gate deletes quiet breathing entirely.

ACOUSTIC PERFORMANCE

- **Transducer type** — analog or PDM MEMS, electret, condenser — and which is recommended for a 24/7 barn.
- **Full frequency response curve with tolerance.**
- **Sensitivity, maximum SPL and Acoustic Overload Point.** The same channel must capture quiet breathing *and* a loud impact without clipping into uselessness.
- **Self-noise (EIN, dB-A) and SNR** for the realistic **mic + preamp + ADC chain**, not the bare transducer datasheet figure.

- **Effective pickup radius** — will it reliably catch quiet breathing and crib-biting from the stall mounting position?
- **Cross-talk between adjacent stalls**, with the polar pattern stated (omni / cardioid / shotgun / array). Attributing one horse's vice to its neighbour is a clinical error.
- **Direction-of-arrival / localisation**, if a multi-mic array is offered.
- **Wind, vibration and structure-borne noise mitigation** — windscreen, shock mount.

DIGITAL AUDIO AND INTERFACING

- **Sample rates and bit depths**, configurable. We require **≥16 kHz**; 32 or 48 kHz preferred.
- **How PCM reaches the edge box** — analog line, **USB Audio (UAC)**, or network.
- **Channel count per device**, and whether multiple mics can be captured **synchronised on a single edge box**.
- **Per-packet metadata** — timestamp, stall/channel ID, gain setting, **clip flag**.
- **End-to-end latency and jitter**, and the buffering / packet size.

ENVIRONMENT, POWER AND PRIVACY

- **IP65+ and an IK (impact) rating**, sealed against **ammonia and urine vapour**, dust and wash-down; operating temperature and humidity, 24/7 rating, MTBF.
- **Power** — PoE (state class and watts), USB or DC, draw per unit.
- **Mounting** for stall ceiling or wall at **2.5–3.5 m**, out of the horse's reach.
- **Encryption in transit (TLS / SRTP) and at rest**, with authentication.
- **DPDP-2023** — stall audio can capture **staff speech**. Confirm audio can be processed and retained **entirely on-site**, with configurable retention and **no undisableable off-site upload**.

7. Edge compute, network and installation — all 12 points

The platform everything else runs on. We supply the software; we need the box, the network, the power and the enclosure specified and installed for Indian barn conditions.

COMPUTE SIZING — THE CENTRAL QUESTION

Per stall the load is **up to 3 × RTSP 1080p @ 25 fps** (IR / visible / fusion) decoded on NVDEC, plus our models, plus IMU, audio and feed/water ingestion.

- **How many stalls per box?** Answer with a **measured benchmark** — GPU %, NVDEC %, CPU %, RAM and thermals — not a theoretical figure.
- **De-rating at temperature** — stalls per box at **45 °C ambient** versus 25 °C.
- **Storage** — NVMe capacity and endurance per box, with **power-loss protection**.
- Platform must be **NVIDIA Jetson (ARM64)**, Ubuntu 22.04+ / current JetPack.

NETWORK AND POWER

- **PoE switch** per barn — per-port and total power budget with **≥25 % headroom**. Cameras, network microphones and any PoE feed/water device all draw from it.
- **Bandwidth** — per-stall and aggregate LAN load at $3 \times 1080p$ @ 25 fps, for H.264 and H.265.
- **4G/LTE failover** — dual-SIM industrial router, Indian carriers and bands, with stated failover behaviour.
- **UPS sizing** — VA/Wh for a stated ride-through, and **clean storage-safe shutdown** with no NVMe corruption.
- **Full local operation during WAN/4G loss** — inference, alerting and buffering must continue with no internet.

PHYSICAL INSTALL AND COMMISSIONING

- **Enclosure** — IP rating, operating temperature, cooling approach (fanless vs filtered), rated for dust, ammonia, humidity, **45–50 °C**, insects/rodents and vibration. A standard office mini-PC enclosure will not survive.
- **Camera mounting** — the horse's head must be in frame at **~3.5 m** with a fixed-focus lens. Mounting geometry, brackets and sight lines.
- **Thermal ↔ visible alignment**. The camera vendor does **not** expose a visible→thermal coordinate mapping. State how you propose to establish and hold that alignment mechanically, since we must relate an ROI in one stream to a region in the other.
- **Zero-touch commissioning** — how a new site is brought up, including stall → camera → horse mapping.

INGESTION OF EACH MODALITY

- **IMU** — BLE gateway hardware, tags per box and per gateway at 100–200 Hz raw, and realistic range through barn structure.
- **Audio** — recommended interface into the Jetson (PoE network mic / USB / I2S).
- **Feed / water** — RS-485 / Modbus, 4–20 mA, pulse and BLE inputs available per box, and how many devices each supports.

FLEET, RESILIENCE AND SECURITY

- **Platform-level health and failure detection** — dead camera, dropped BLE gateway, failed sensor. Silence must be detectable as silence.
- **Time discipline** — NTP source or PTP grandmaster, and how the box's clock is held accurate for the cross-modality alignment in G6.
- **Remote update and recovery** — how OS and firmware are updated and how a bricked box is recovered, for a handful of boxes at one site.
- **Edge security** — full-disk encryption, secure boot, TPM or secure-element key storage; how per-device credentials are provisioned and rotated.
- **Data residency** — all processing and storage on-premise in India. Any cloud sync only to an **India region** (e.g. GCP asia-south1), including OTA and telemetry.

DEPLOYMENT BOUNDARY

- **Deployment unit** — per barn, or per N stalls? Provide a **per-site bill of materials**.
- **Integration boundary** — state precisely what you deliver versus what BSV delivers, and what support looks like after handover.

8. Evaluation hardware

We validate on our own bench before committing to a design. For each track we would like evaluation units together with the **SDK/driver, protocol documentation and sample code** for Linux/ARM64:

Track	Evaluation units requested
IMU	3–5 tags + 1 gateway + charging dock
Feed / water	1 water sensor + 1 weigh-back feeder
Microphone	2 microphones + any required interface or PoE injector
Edge infrastructure	1 edge unit configured for the pilot, with switch, router and UPS

Please state availability and lead time.

9. What we need attached with a reply

So that we can evaluate in a single pass, without rounds of follow-up:

- Every requirement explicitly **confirmed or declined** — "not supported" is a perfectly good answer and costs nothing. A blank or skipped item we have to record as *not supported*, because we cannot tell the difference.
- Sensor and component **datasheets**.
- The **protocol document, register map or API reference** — the actual file.
- An **ARM64 (aarch64) build**, attached or as a download link — *or* a plain statement that none exists.
- A **sample data payload** showing real field names and units.
- Sample code, if any is provided.
- Evaluation-unit availability and lead time.
- A named technical contact we can reach directly.

We would far rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

Commercial terms are not part of this document — those are discussed separately once technical fit is confirmed.